Homework 4.2: An Explicit JKO Step for the Heat Equation

Introduction

In this homework, I study the Jordan-Kinderlehrer-Otto (JKO) scheme for the heat equation.

Consider the entropy functional

$$\mathcal{E}(\rho) = \int_{\mathbb{R}} \rho(x) \log \rho(x)\, dx.$$

Its Wasserstein gradient flow is the heat equation

$$\partial_t \rho = \partial_{xx} \rho.$$

Assume that the initial density is a centered Gaussian

$$\rho^0 = \mathcal{N}(0, \sigma_0^2).$$

Since the heat equation preserves Gaussianity, I restrict the JKO scheme to centered Gaussian densities

$$\rho = \mathcal{N}(0, \sigma^2).$$

This reduces the infinite-dimensional Wasserstein minimization problem to a scalar optimization problem involving the standard deviation σ.

The main goals of this homework are to:

- compute the entropy of a centered Gaussian;
- compute the Wasserstein distance between centered Gaussians;
- derive an explicit JKO update for σ;
- compare the discrete JKO evolution with the exact heat equation;
- investigate convergence as the time step τ decreases;
- evaluate the mathematical reliability of an LLM-generated solution.

Part I: Entropy of a Gaussian

Let

$$\rho = \mathcal{N}(0, \sigma^2).$$

Its density is

$$\rho(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

The entropy functional used in the JKO scheme is

$$\mathcal{E}(\rho) = \int_{\mathbb{R}} \rho(x) \log \rho(x)\, dx.$$

Derivation

Taking the logarithm of the Gaussian density gives

$$\log \rho(x) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{x^2}{2\sigma^2}.$$

Therefore,

$$\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi\sigma^2) \int_{\mathbb{R}} \rho(x)\, dx - \frac{1}{2\sigma^2} \int_{\mathbb{R}} x^2 \rho(x)\, dx.$$

Since ρ is a probability density,

$$\int_{\mathbb{R}} \rho(x)\, dx = 1.$$

Also,

$$\int_{\mathbb{R}} x^2 \rho(x)\, dx = \sigma^2.$$

Thus,

$$\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{1}{2}.$$

Since

$$\frac{1}{2} = \frac{1}{2}\log e,$$

we obtain

$$\boxed{\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e\sigma^2)}.$$

Expanding the logarithm gives

$$\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e) - \log \sigma.$$

Therefore, up to constants independent of σ,

$$\boxed{\mathcal{E}(\rho) = -\log \sigma.}$$

Numerical Verification

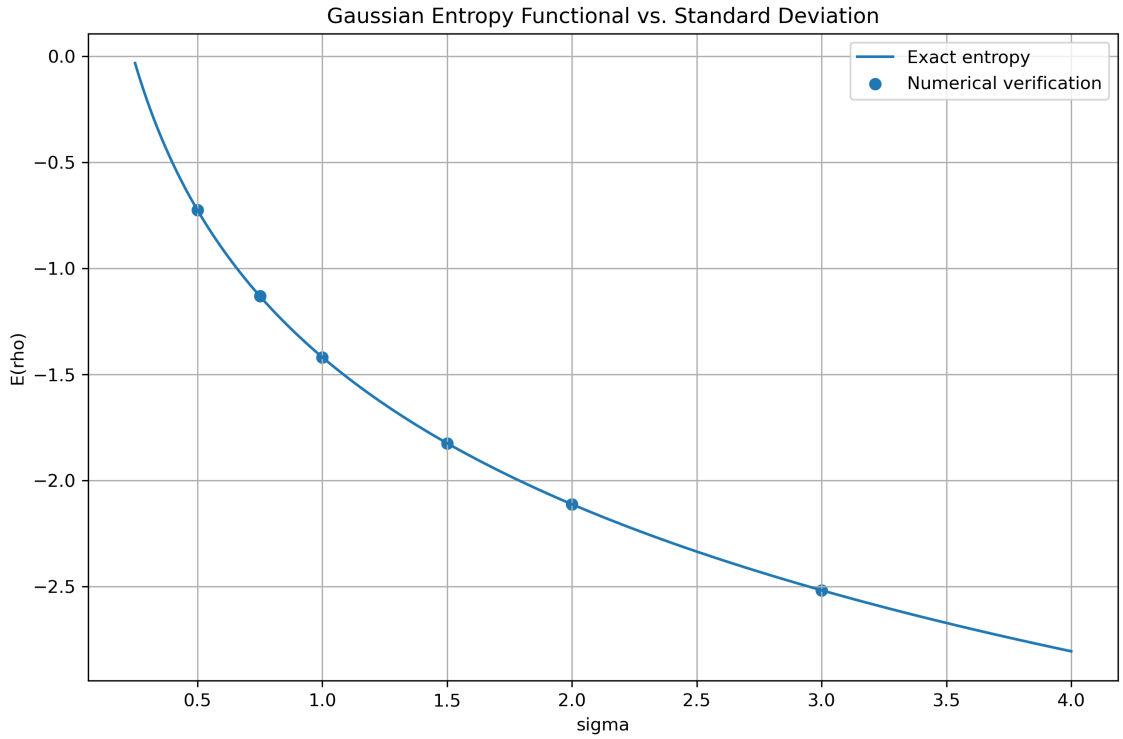
The entropy formula was verified numerically for several values of σ.

For each Gaussian, I checked:

- normalization;
- the second moment;
- the full entropy functional;
- the reduced entropy after removing the constant independent of σ.

The numerical entropy values lie on the theoretical curve

$$-\frac{1}{2}\log(2\pi e\sigma^2).$$

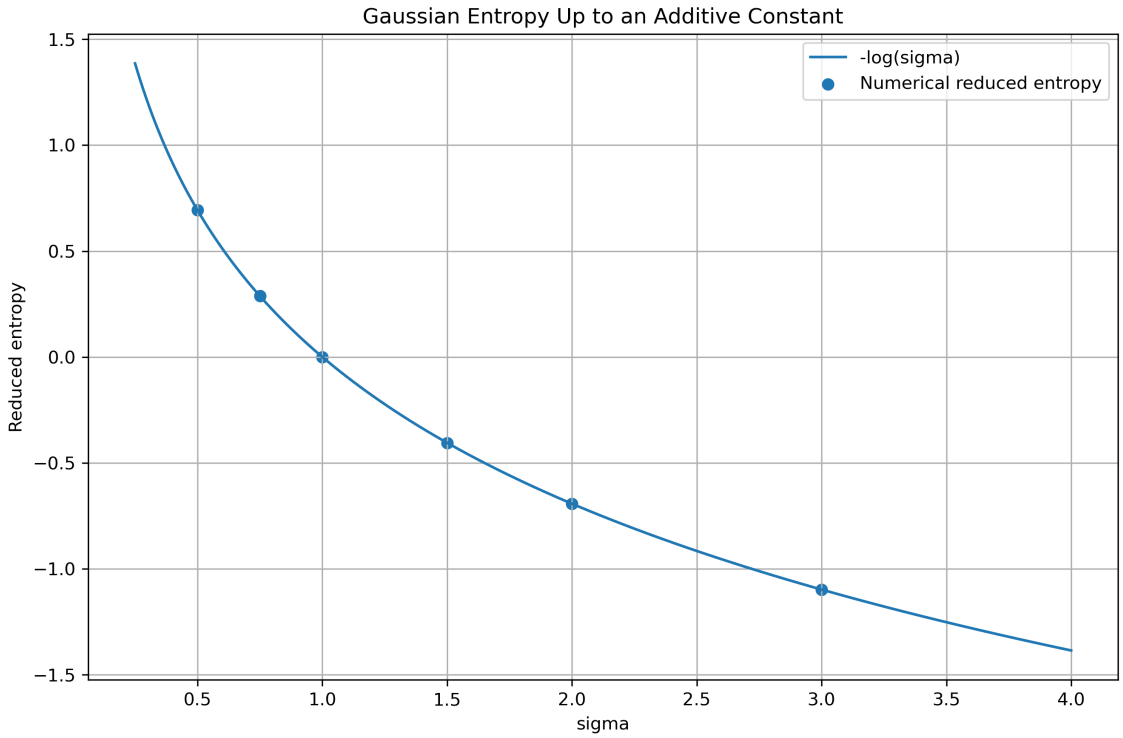


After removing the constant

$$-\frac{1}{2}\log(2\pi e),$$

the remaining quantity agrees with

$$-\log \sigma.$$



Since

$$\frac{d\mathcal{E}}{d\sigma} = -\frac{1}{\sigma} < 0,$$

the functional decreases as the Gaussian spreads.

The functional

$$\int \rho \log \rho \, dx$$

is the negative of the usual differential entropy.

Part II: Wasserstein Distance Between Centered Gaussians

Consider

$$\mu = \mathcal{N}(0, \sigma^2)$$

and

$$\nu = \mathcal{N}(0, \sigma_k^2).$$

The goal is to compute

$$W_2^2(\mu, \nu).$$

Quantile Derivation

In one dimension,

$$W_2^2(\mu, \nu) = \int_0^1 \left|F_\mu^{-1}(u) - F_\nu^{-1}(u)\right|^2 du.$$

Let Φ denote the standard Gaussian CDF.

For a centered Gaussian,

$$F_\mu^{-1}(u) = \sigma \Phi^{-1}(u)$$

and

$$F_\nu^{-1}(u) = \sigma_k \Phi^{-1}(u).$$

Therefore,

$$W_2^2(\mu, \nu) = (\sigma - \sigma_k)^2 \int_0^1 [\Phi^{-1}(u)]^2 du.$$

If

$$U \sim \text{Uniform}(0, 1),$$

then

$$\Phi^{-1}(U) \sim \mathcal{N}(0, 1).$$

Hence,

$$\int_0^1 [\Phi^{-1}(u)]^2 du = 1.$$

Thus,

$$\boxed{W_2^2\left(\mathcal{N}(0, \sigma^2), \mathcal{N}(0, \sigma_k^2)\right) = (\sigma - \sigma_k)^2.}$$

Equivalently,

$$\boxed{W_2 = |\sigma - \sigma_k|.}$$

The formula depends on the difference between the standard deviations, not the difference between the variances.

Optimal Transport Map

The monotone optimal map from

$$\mathcal{N}(0, \sigma_k^2)$$

to

$$\mathcal{N}(0, \sigma^2)$$

is

$$T(x) = \frac{\sigma}{\sigma_k} x.$$

Then

$$W_2^2 = \int_{\mathbb{R}} |T(x) - x|^2 \rho_k(x)\, dx.$$

Since

$$T(x) - x = \left(\frac{\sigma}{\sigma_k} - 1 \right) x,$$

we get

$$W_2^2 = \left(\frac{\sigma - \sigma_k}{\sigma_k} \right)^2 \int_{\mathbb{R}} x^2 \rho_k(x)\, dx.$$

Using

$$\int_{\mathbb{R}} x^2 \rho_k(x)\, dx = \sigma_k^2,$$

we again obtain

$$\boxed{W_2^2 = (\sigma - \sigma_k)^2.}$$

Numerical Verification

The formula was checked numerically using both:

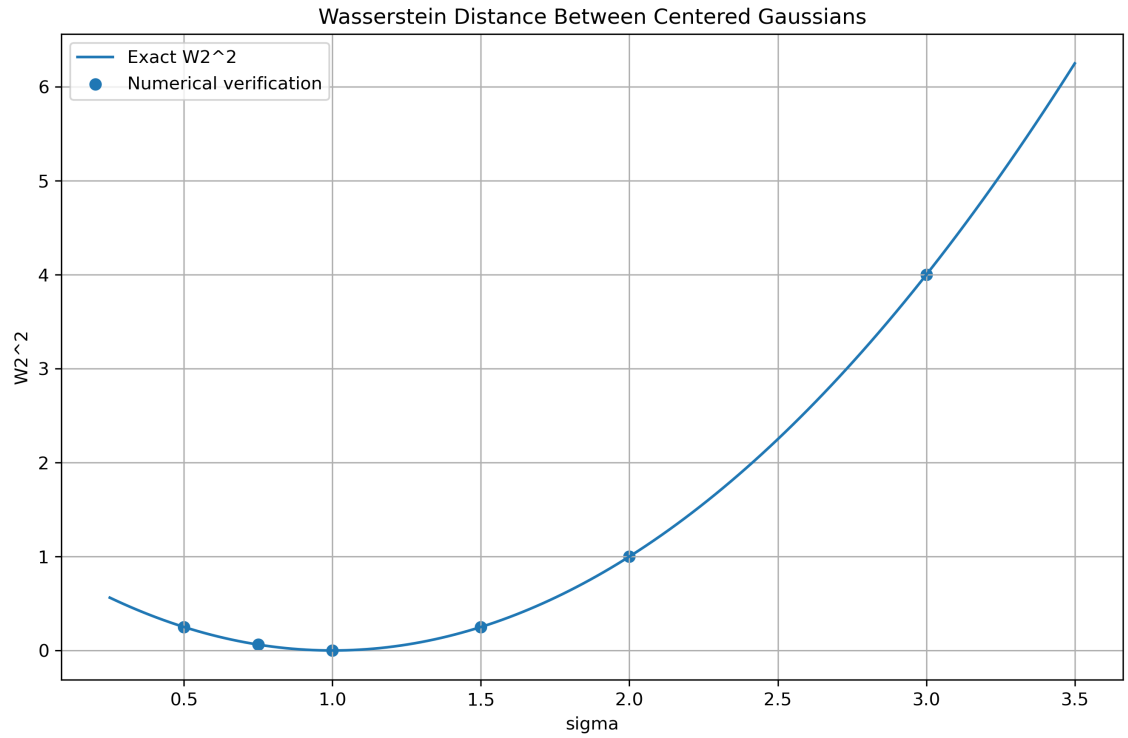
- the optimal transport map;
- the quantile representation.

For fixed

$$\sigma_k = 1,$$

the theoretical curve is

$$W_2^2 = (\sigma - 1)^2.$$



The numerical values lie directly on the theoretical parabola.

Thus the numerical computation confirms

$$\boxed{W_2^2 = (\sigma - \sigma_k)^2.}$$

Part III: The JKO Minimization

From Parts I and II, the JKO step becomes

$$\sigma_{k+1} = \arg\min_{\sigma>0} \left\{ -\log\sigma + \frac{1}{2\tau}(\sigma - \sigma_k)^2 \right\}.$$

Define

$$J(\sigma) = -\log\sigma + \frac{1}{2\tau}(\sigma - \sigma_k)^2.$$

First-Order Condition

Differentiate:

$$J'(\sigma) = -\frac{1}{\sigma} + \frac{\sigma - \sigma_k}{\tau}.$$

At the minimizer,

$$J'(\sigma) = 0.$$

Therefore,

$$-\frac{1}{\sigma} + \frac{\sigma - \sigma_k}{\tau} = 0.$$

Multiplying by $\tau\sigma$ gives

$$\sigma(\sigma - \sigma_k) = \tau.$$

Thus,

$$\sigma^2 - \sigma_k\sigma - \tau = 0.$$

Using the quadratic formula,

$$\sigma = \frac{\sigma_k \pm \sqrt{\sigma_k^2 + 4\tau}}{2}.$$

Because

$$\sqrt{\sigma_k^2 + 4\tau} > \sigma_k,$$

the negative root satisfies

$$\frac{\sigma_k - \sqrt{\sigma_k^2 + 4\tau}}{2} < 0.$$

Since σ is a standard deviation,

$$\sigma > 0.$$

Therefore, the negative root must be discarded.

The JKO update is

$$\boxed{\sigma_{k+1} = \frac{\sigma_k + \sqrt{\sigma_k^2 + 4\tau}}{2}.}$$

Convexity

The second derivative is

$$J''(\sigma) = \frac{1}{\sigma^2} + \frac{1}{\tau}.$$

For

$$\sigma > 0, \qquad \tau > 0,$$

we have

$$J''(\sigma) > 0.$$

Therefore, the objective is strictly convex, and the positive critical point is the unique global minimizer.

Numerical Verification

The explicit formula was compared with an independent numerical optimization for several values of σ_k and τ.

The maximum difference between the explicit formula and the numerical optimizer was

$$2.39 \times 10^{-9}.$$

The largest value of

$$|J'(\sigma_{k+1})|$$

was

$$1.45 \times 10^{-14}.$$

The maximum residual in

$$\sigma_{k+1}^2 - \sigma_k \sigma_{k+1} - \tau = 0$$

was

$$5.27 \times 10^{-16}.$$

The smallest computed value of

$$J''(\sigma_{k+1})$$

was

$$1.09167 > 0.$$

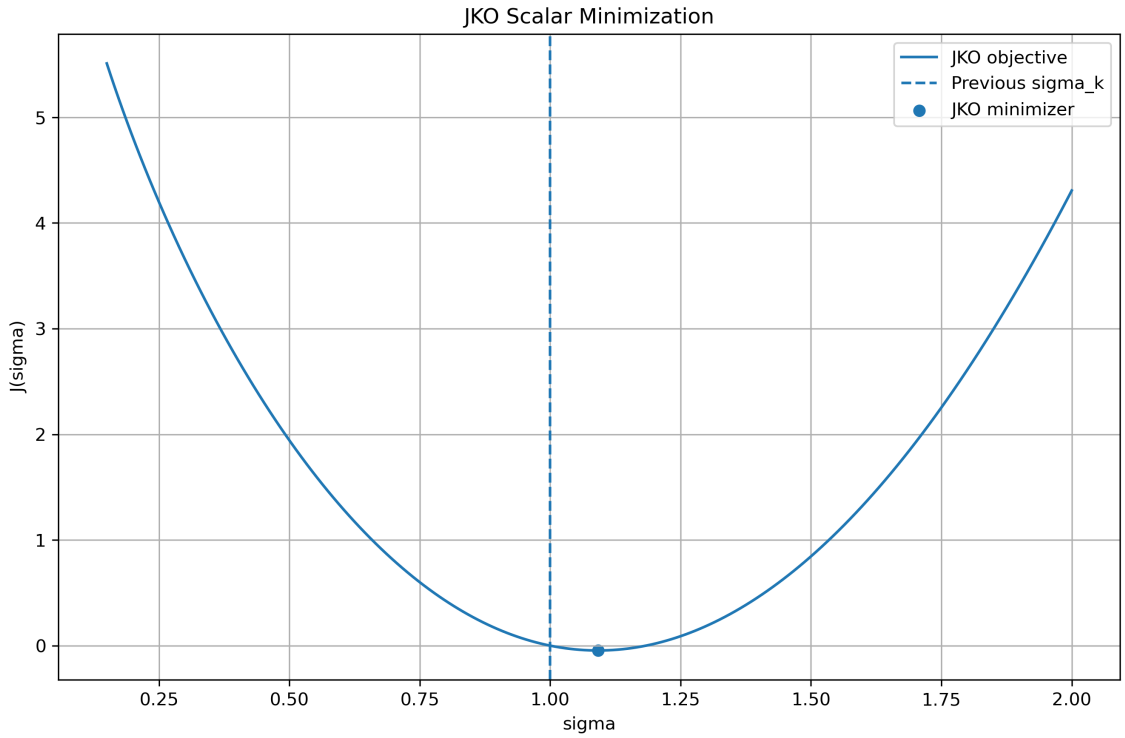
Thus the analytic minimizer is strongly verified.

For the example

$$\sigma_k = 1, \qquad \tau = 0.1,$$

the minimizer is

$$\sigma_{k+1} \approx 1.091608.$$

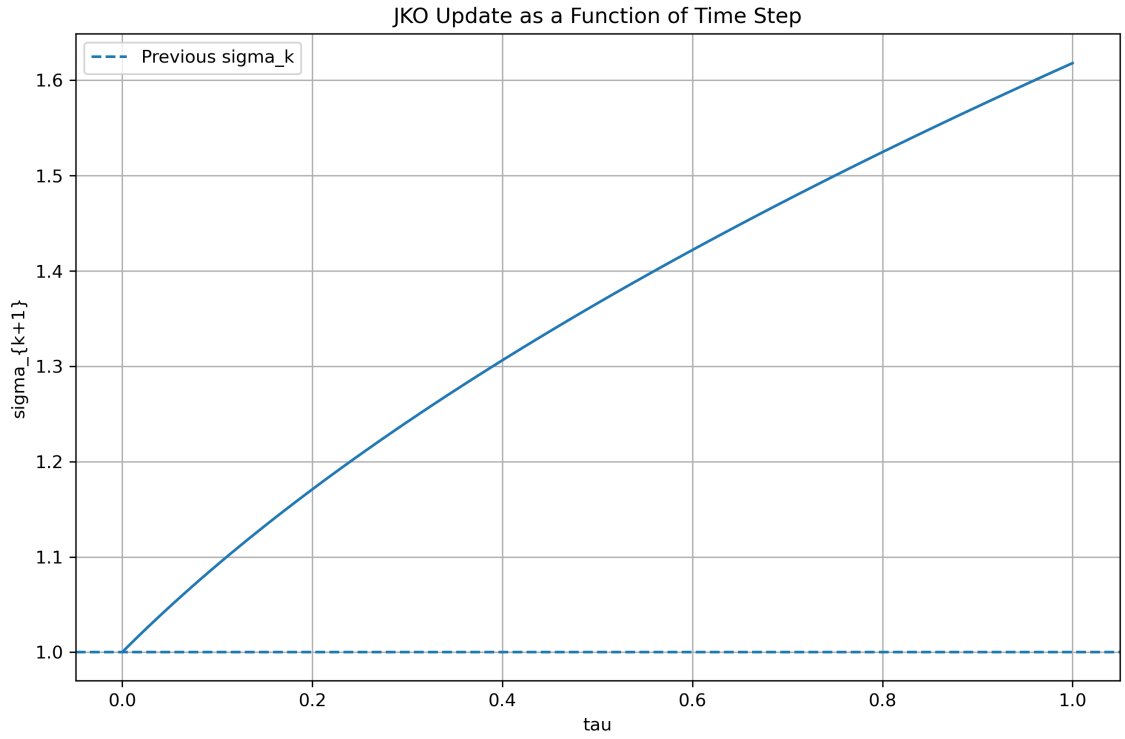


The minimum lies to the right of

$$\sigma_k = 1,$$

showing that the Gaussian spreads after one JKO step.

For fixed $\sigma_k = 1$, the update increases as τ increases.



The optimality condition can also be written as

$$\boxed{\sigma_{k+1} - \sigma_k = \frac{\tau}{\sigma_{k+1}}.}$$

Thus every JKO step satisfies

$$\sigma_{k+1} > \sigma_k.$$

Part IV: Compare with the Heat Equation

The exact heat equation solution starting from

$$\rho^0 = \mathcal{N}(0, \sigma_0^2)$$

is

$$\rho(t) = \mathcal{N}\left(0, \sigma_0^2 + 2t\right).$$

Therefore,

$$\boxed{\sigma^2_{\text{exact}}(t) = \sigma_0^2 + 2t.}$$

At the JKO time levels

$$t_k = k\tau,$$

the exact variance is

$$\boxed{\sigma_0^2 + 2k\tau.}$$

Variance Increment of the JKO Scheme

From Part III,

$$\sigma_{k+1} - \sigma_k = \frac{\tau}{\sigma_{k+1}}.$$

Then

$$\sigma_{k+1}^2 - \sigma_k^2 = (\sigma_{k+1} - \sigma_k)(\sigma_{k+1} + \sigma_k).$$

Therefore,

$$\sigma_{k+1}^2 - \sigma_k^2 = \frac{\tau}{\sigma_{k+1}}(\sigma_{k+1} + \sigma_k).$$

Since

$$\sigma_k = \sigma_{k+1} - \frac{\tau}{\sigma_{k+1}},$$

we obtain

$$\boxed{\sigma_{k+1}^2 - \sigma_k^2 = 2\tau - \frac{\tau^2}{\sigma_{k+1}^2}.}$$

The exact heat equation increases the variance by

$$2\tau.$$

Therefore,

$$\sigma_{k+1}^2 - \sigma_k^2 < 2\tau.$$

This shows that the finite-step JKO scheme slightly underestimates the exact variance growth.

Numerical Comparison

For the numerical experiment, I chose

$$\sigma_0 = 1$$

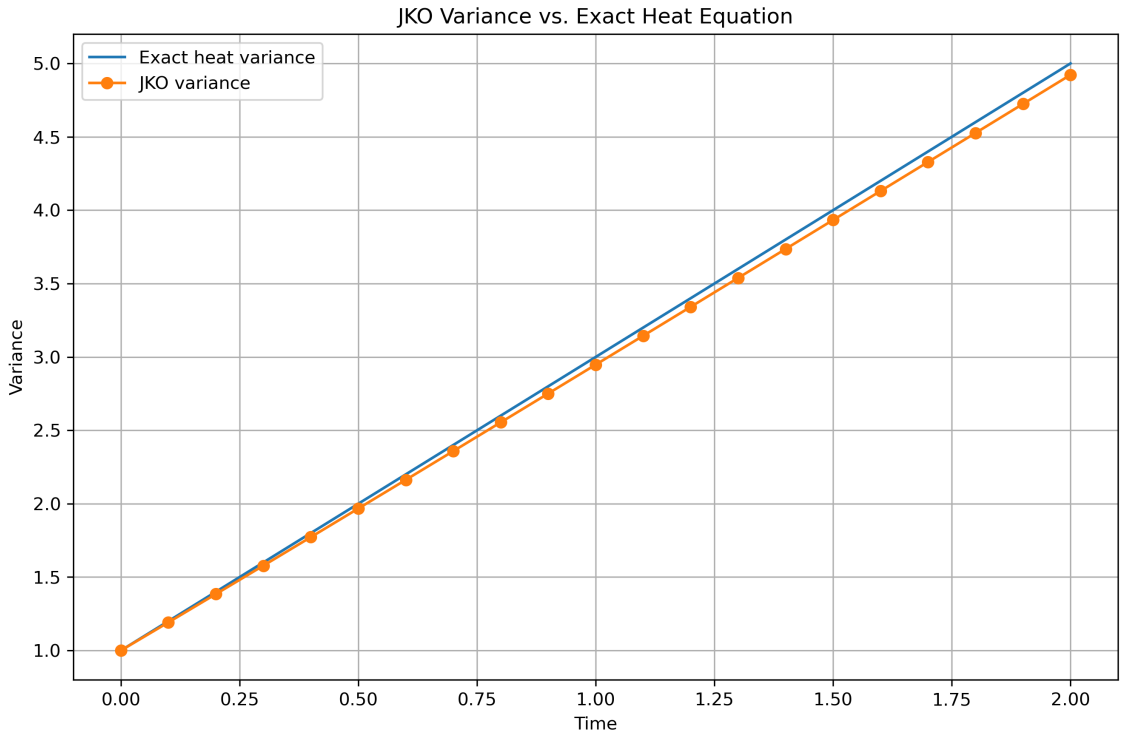
and

$$T = 2.$$

For

$$\tau = 0.1,$$

the JKO variance closely follows the exact heat equation.



The JKO curve remains slightly below the exact variance, as predicted analytically.

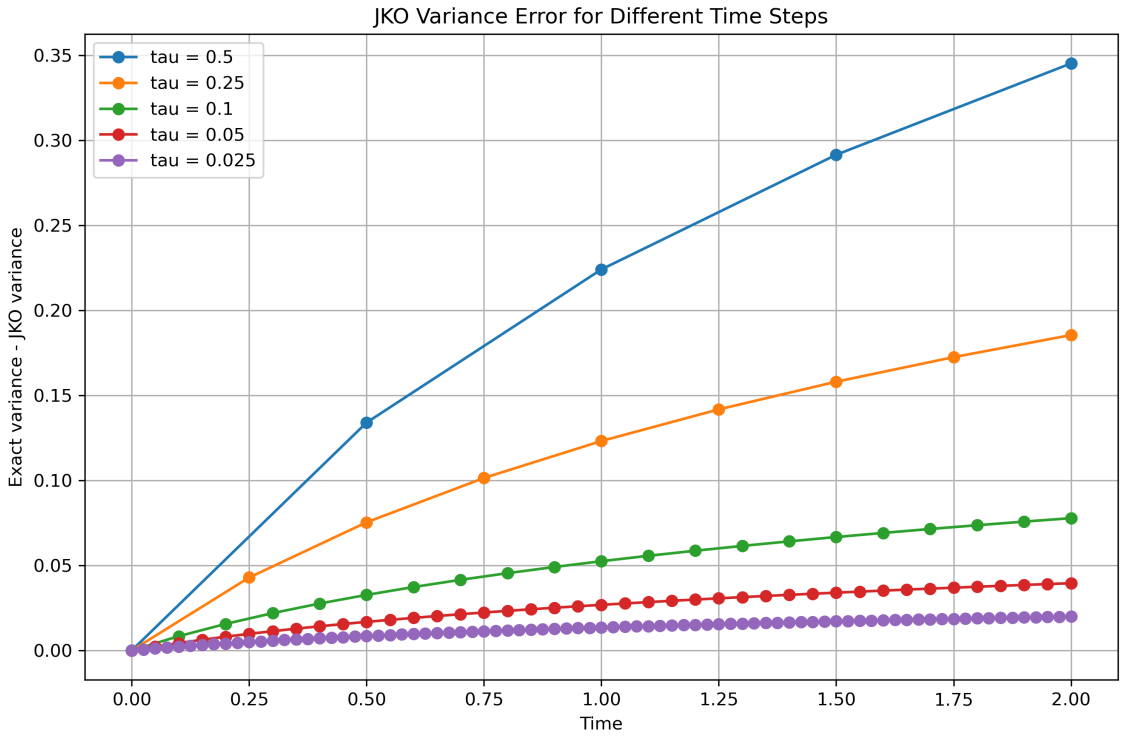
The experiment was repeated for

$$\tau = 0.5,\ 0.25,\ 0.1,\ 0.05,\ 0.025.$$

The final-time results were:

τ	Steps	JKO variance	Exact variance	Error
0.500	4	4.654848	5.000000	0.345152
0.250	8	4.814548	5.000000	0.185452
0.100	20	4.922209	5.000000	0.077791
0.050	40	4.960449	5.000000	0.039551
0.025	80	4.980055	5.000000	0.019945

The error decreases consistently as τ becomes smaller.



The numerical variance-increment identity was verified with maximum residual

$$2.11 \times 10^{-15}.$$

Convergence Rate

The observed convergence orders were

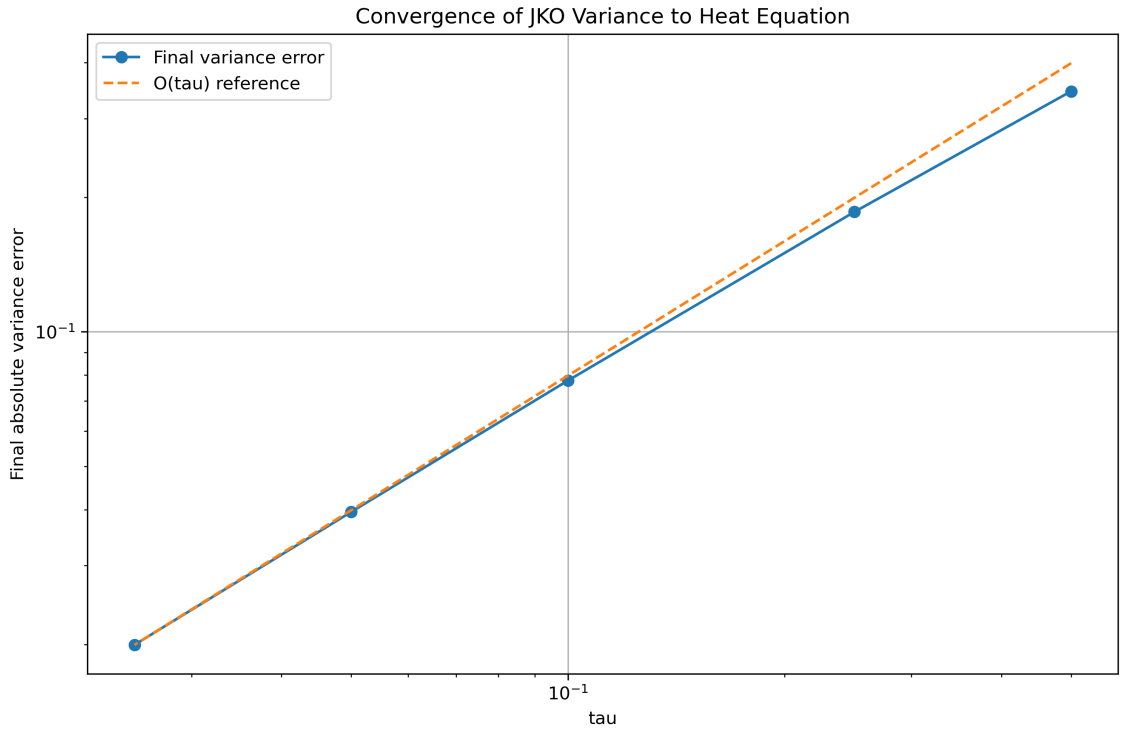
$$0.896, \qquad 0.948, \qquad 0.976, \qquad 0.988.$$

These approach

$$1$$

as

$$\tau \to 0.$$



Therefore, the numerical experiment indicates approximately first-order convergence:

$$\boxed{\text{error} = O(\tau).}$$

This is also consistent with the optimality condition

$$\frac{\sigma_{k+1} - \sigma_k}{\tau} = \frac{1}{\sigma_{k+1}},$$

which is the backward Euler discretization of

$$\frac{d\sigma}{dt} = \frac{1}{\sigma}.$$

Multiplying by 2σ gives

$$\frac{d}{dt}\sigma^2 = 2,$$

so

$$\sigma^2(t) = \sigma_0^2 + 2t.$$

Thus the JKO iteration converges to the exact Gaussian heat flow as

$$\tau \to 0.$$

Part V: LLM Critique

Prompt Given to the LLM

> Solve Homework 4.2 for a centered Gaussian.
>
> Compute the Gaussian entropy functional, derive the Wasserstein distance between centered Gaussians, solve the scalar JKO minimization explicitly, and compare the resulting iteration with the exact heat equation.
>
> Be careful to distinguish between the standard deviation σ and variance σ^2.

The LLM-generated response was checked according to the criteria in the assignment.

1. Gaussian Entropy

The LLM correctly obtained

$$\boxed{\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e\sigma^2)}.$$

It also correctly simplified this, up to a constant independent of σ, to

$$\boxed{\mathcal{E}(\rho) = -\log\sigma.}$$

This agrees with both the analytic derivation and numerical verification from Part I.

Therefore, the Gaussian entropy calculation is correct.

2. Wasserstein Distance

The LLM correctly used the one-dimensional quantile formula and obtained

$$\boxed{W_2^2\left(\mathcal{N}(0,\sigma^2),\mathcal{N}(0,\sigma_k^2)\right) = (\sigma-\sigma_k)^2.}$$

This agrees with the independent optimal-map and quantile calculations from Part II.

Therefore, the Wasserstein calculation is correct.

3. Positive Root for σ_{k+1}

The LLM correctly derived

$$\sigma^2 - \sigma_k\sigma - \tau = 0.$$

It then obtained

$$\sigma = \frac{\sigma_k \pm \sqrt{\sigma_k^2 + 4\tau}}{2}.$$

The negative root is less than zero and therefore cannot represent a standard deviation.

The LLM correctly selected

$$\boxed{\sigma_{k+1} = \frac{\sigma_k + \sqrt{\sigma_k^2 + 4\tau}}{2}.}$$

It also checked

$$J''(\sigma) > 0,$$

showing that this is the unique minimizer.

Therefore, the JKO minimization is correct.

4. Distinction Between σ and σ^2

The LLM does not confuse the standard deviation

$$\sigma$$

with the variance

$$\sigma^2.$$

It correctly uses

$$(\sigma - \sigma_k)^2$$

for the squared Wasserstein distance.

It then correctly compares

$$\sigma_k^2$$

with the exact heat-equation variance

$$\sigma_0^2 + 2k\tau.$$

Thus the distinction between standard deviation and variance is maintained throughout the derivation.

5. Sign and Behavior of the Entropy

The LLM correctly observes that

$$\frac{d\mathcal{E}}{d\sigma} = -\frac{1}{\sigma} < 0.$$

Therefore, the functional decreases as the Gaussian spreads.

The LLM also correctly notes that

$$\int \rho \log \rho \, dx$$

is the negative of the usual differential entropy.

One useful clarification made by the LLM is that this functional is not negative for every possible value of σ.

Indeed,

$$\mathcal{E}(\rho) < 0$$

when

$$2\pi e \sigma^2 > 1.$$

For sufficiently small σ, the functional can be positive.

Thus the most important general statement is not that it is always negative, but rather that

$$\boxed{\mathcal{E}(\rho) \text{ decreases as } \sigma \text{ increases.}}$$

Minor Issue in the LLM Response

The convergence statement

$$\sigma_k^2 \to \sigma_0^2 + 2k\tau \qquad \text{as } \tau \to 0$$

should be interpreted more carefully.

When taking

$$\tau \to 0,$$

the number of steps k should increase so that the physical time

$$t = k\tau$$

remains fixed.

A more precise statement is:

$$k\tau \to t$$

and

$$\sigma_k^2 \to \sigma_0^2 + 2t.$$

This is a minor issue of mathematical precision and does not affect the main derivation.

Overall Assessment of the LLM

The LLM performed well on this problem.

It correctly:

- computed the Gaussian entropy;
- computed the Wasserstein distance;
- derived the JKO minimizer;
- selected the physically meaningful positive root;
- distinguished standard deviation from variance;
- recognized that the entropy functional decreases as the Gaussian spreads;
- connected the JKO iteration to the exact heat equation.

The main formulas were also independently verified numerically.

Therefore, the LLM was mathematically reliable for this problem, although the convergence statement benefited from a small clarification concerning the limit

$$\tau \to 0.$$

Conclusion

For a centered Gaussian

$$\rho = \mathcal{N}(0, \sigma^2),$$

the entropy functional is

$$\boxed{\mathcal{E}(\rho) = -\frac{1}{2}\log(2\pi e \sigma^2)}.$$

Up to constants independent of σ,

$$\boxed{\mathcal{E}(\rho) = -\log \sigma.}$$

The squared Wasserstein distance between two centered Gaussians is

$$\boxed{W_2^2\left(\mathcal{N}(0, \sigma^2), \mathcal{N}(0, \sigma_k^2)\right) = (\sigma - \sigma_k)^2.}$$

The JKO minimization therefore reduces to

$$\arg\min_{\sigma>0}\left\{-\log\sigma+\frac{1}{2\tau}(\sigma-\sigma_k)^2\right\},$$

whose unique minimizer is

$$\boxed{\sigma_{k+1}=\frac{\sigma_k+\sqrt{\sigma_k^2+4\tau}}{2}.}$$

The exact heat equation satisfies

$$\boxed{\sigma^2(t)=\sigma_0^2+2t.}$$

For finite τ, the JKO variance grows slightly more slowly than the exact heat-equation variance.

However, the numerical experiments show that the error decreases approximately linearly with τ.

The observed convergence orders approach

$$1,$$

confirming approximately first-order convergence.

Therefore, the explicit Gaussian JKO scheme provides a consistent discrete approximation of the Wasserstein gradient flow of entropy, and in the limit

$$\tau\to 0,$$

it reproduces the exact Gaussian solution of the heat equation.